

Chemistry

(A Level, Edexcel)

This course teaches the theoretical and practical skills necessary to become an advanced chemist. There will normally be one or two practical sessions per week backed up by theoretical work. The course is designed to relate to real-world chemistry and each topic is framed by the context in which the chemistry can be applied. Our course still involves pops and squeaks, colour changes and smells. You will gain new skills which are useful in the subject, but many are also transferrable. Chemistry is a subject of endless possibilities; many of our ex-students are involved in cutting edge research into new drugs and materials. In order to do well on this course, you will need to work very hard outside the classroom, solving problems and assimilating new ideas.

Entry Requirements:

In Combined (Double) Science: Grade 6 in both exams.

In Separate (Triple) Sciences: Grade 6 in GCSE Chemistry, plus a 6 in another Science GCSE.

Grade 6 or above in GCSE Mathematics.

Course Content

A-level Topics in Year 12

Topic 1 Atomic structure and the periodic table
Topic 2 Bonding and structure
Topic 3 Redox 1
Topic 4 Inorganic Chemistry and the Periodic Table
Topic 5 Formulae equations and Amounts of Substance
Topic 6 Organic Chemistry 1
Topic 7 Modern Analytical Techniques 1
Topic 8 Energetics
Topic 9 Kinetics 1
Topic 10 Equilibrium 1

A-level Topics in Year 13

Topic 11 Equilibrium 2
Topic 12 Acid-base Equilibria
Topic 13 Energetics
Topic 14 Redox 2
Topic 15 Transition Metals
Topic 16 Kinetics 2
Topic 17 Organic Chemistry 2
Topic 18 Organic Chemistry 3
Topic 19 Modern Analytical Techniques 2

Assessment

Assessment consists of three exams at the end of Year 13, Papers 1 and 2 cover separate topics with Paper 3 being a synoptic paper which has a stronger focus on the practical elements of the course.

There is also a practical endorsement. Course endorsed practical skills are assessed continuously by the teacher(s), including during 16 Core Practical assessments throughout Years 12 and 13. Only at the completion of the course is a pass or fail assessment submitted to the exam board.

Career Links

Students who have studied Chemistry at Backwell have gone to university to study for degrees in Chemistry, Biochemistry, Natural Sciences, Medicine, Veterinary Medicine, Pharmacy, Pharmacology, Environmental Sciences and various Engineering degrees. A level Chemistry is a qualification that is highly valued by universities and employers as they demonstrate abilities in analysis and problem solving which are sought after skills.

For more information please contact Dr A Hancock